Reliable Smartphone-Based Detection of Counterfeit Bangladeshi Banknotes: Multi-View Deep Learning with Out-of-Distribution Rejection and a Benchmark Audit

Debobrato Biswas ORCID[1*] and Imran Mahmud[1]

[1*]Department of Software Engineering, Daffodil International University, Dhaka, 1207, Bangladesh.

*Corresponding author(s). E-mail(s): biswas2305343003@diu.edu.bd;

Abstract

Background: Counterfeit high-value banknotes harm retailers and individuals who cannot afford verification hardware.

Methods: We present a smartphone-based authenticator for Bangladeshi Tk 500 and Tk 1000 notes, evaluated on JaalTaka (1,390 notes: 802 genuine, 588 seized counterfeits; six photographs each). It fine-tunes EfficientNet-B0, fuses up to six photographs per note, returns a genuine, uncertain, or counterfeit verdict, rejects unsupported inputs, and explains decisions with Grad-CAM. We audited the benchmark for duplicates.

Results: On 208 held-out test notes, the system reaches 98.08% accuracy (95% CI 95.16–99.25%) and an ROC-AUC of 0.994 at 37 ms per photograph. However, the 1,390 notes form only 846 distinct photograph sets, 116 test notes have a copy in training or validation, and 12 duplicate groups carry both labels, causing all four test errors. On the 92 test notes without a copy, the system and two simple baselines are error-free. On a group-aware split without shared photographs, the retrained system and nearest-neighbour classification on generic features are both error-free on all 205 test notes.

Conclusion: The benchmark is saturated. We release the duplicate groups and recommend group-aware, counterfeit-family hold-out evaluation.

Keywords: Counterfeit banknote detection, Bangladeshi Taka, Benchmark audit, Data leakage, Out-of-distribution detection, Explainable AI

Plain Language Summary

Fake Tk 500 and Tk 1000 notes are hard to spot by eye. We built a tool that checks a banknote from one to six phone photographs, answers “genuine”, “counterfeit”, or “uncertain”, refuses pictures of other objects, and highlights the parts of the note behind its answer. It was correct for 98% of the test notes of a public photo collection. That collection, however, contains many repeated photographs, some labelled both genuine and fake, and these explained every mistake. The tool may therefore not catch new kinds of fakes. We share the repeated photographs and suggest fairer tests.

Word Count

3,958 words excluding references (main text: 2,922 words; abstract: 183 words); 9 figures; 6 tables.

1 Introduction

Whoever accepts a counterfeit Tk 500 or Tk 1000 note bears its full value as a loss. Banks use ultraviolet, infrared, and magnetic sorters, but shopkeepers and citizens rely on visual and tactile checks. Authentication with an ordinary smartphone camera is an attractive complement and has been studied for other currencies [1, 2].

The JaalTaka dataset [3] enabled such work for Bangladeshi notes by releasing smartphone photographs of genuine notes and of counterfeits seized by the Rapid Action Battalion (RAB); its authors reported 97.4–98.9% accuracy for five ImageNet CNNs. A tool trusted with money must also (i) handle unsupported inputs, (ii) admit uncertainty, (iii) explain its decisions, and (iv) be evaluated in a way that predicts behaviour on unseen notes. The last point is easy to get wrong: networks exploit shortcuts [4], and duplicates across splits inflate accuracy [5–7].

We ask two research questions. *RQ1:* Does the near-perfect accuracy reported on JaalTaka measure counterfeit recognition or properties of the benchmark? *RQ2:* Can a smartphone authenticator meet (i)–(iii) in real time on a CPU without losing accuracy? Our contributions are:

1. a duplicate audit of JaalTaka, to our knowledge the first for a counterfeit-banknote benchmark, showing that its 1,390 notes form only 846 distinct photograph sets, that label conflicts cause every test error, and that nearest-neighbour classification on generic features already saturates it; we release the duplicate groups and re-evaluate on a group-aware split;
2. a rigorous evaluation protocol with note-disjoint and group-aware splits, group-level bootstrap confidence intervals, paired significance tests, frozen-feature baselines, and an occlusion test that reveals signal outside the note's centre; and
3. an open, CPU-real-time pipeline (Fig. 1) combining EfficientNet-B0 [8], fusion of up to six photographs, a three-way decision with adjustable strictness, an out-of-distribution (OOD) gate, and Grad-CAM [9].

2 Related Work

Hand-crafted counterfeit detection.

Early detectors used luminance histograms with multiple-kernel SVMs [10], the texture of scanned background printing [11], the multispectral response of inks [12], and individual security features of Bangladeshi notes [13]; a mobile application was also proposed [14]. These methods need controlled imaging (scanners or multispectral sensors) and denomination-specific features, which limits their use with phone cameras.

Deep learning and Bangladeshi notes.

Transfer learning from ImageNet [15, 16] lets CNNs learn from few images. Pham et al. detected fake banknotes in visible-light smartphone images [1] and tested their models across currencies [2]. For Bangladeshi notes, denominations were recognised with neural networks and lightweight CNNs [17, 18], and the BanglaTaka dataset [19] targets denomination classification rather than authentication. Counterfeit detection became possible at scale with JaalTaka [3], which was benchmarked with DenseNet121 [20], ResNet50 [21], VGG16 [22], MobileNetV2 [23], and InceptionV3 [24], and transfer learning was also applied to Bangladeshi counterfeits [25]. These studies compare architectures by test accuracy on random splits; none reports OOD rejection, an uncertain outcome, or a duplicate check (Table 1), so a high score cannot be separated from memorisation of near-identical images.

Reliable prediction.

Maximum softmax probability is a weak OOD signal [26]. Stronger post-hoc detectors use Mahalanobis distances on network features [27], energy scores [28], or nearest-neighbour distances [29]; we adapt the Mahalanobis approach to a low-dimensional subspace because it needs only the training embeddings and is cheap on a CPU. Miscalibrated probabilities [30] motivate an explicit "uncertain" outcome, and Grad-CAM [9] shows the regions supporting a class.

Leakage and benchmark audits.

Information shared between training and test data inflates reported performance [5, 7]. Barz and Denzler found near-duplicates of test images

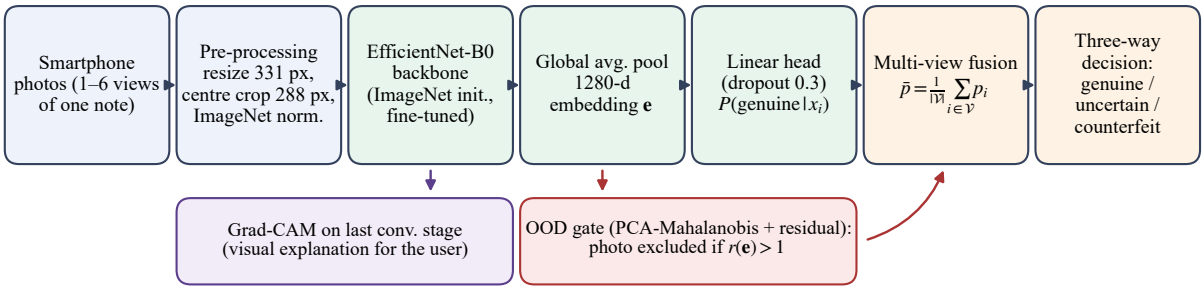


Fig. 1 System overview: per-photograph EfficientNet-B0 classification, OOD filtering, averaging, and a three-way verdict with Grad-CAM maps

in the training sets of CIFAR and showed that removing them changes model rankings [6], and grouped data require splits that keep each group in one partition [31]. Shortcut learning [4] adds a related risk: a network can succeed by exploiting acquisition artefacts. We apply these ideas to a counterfeit benchmark, where augmented copies were distributed as separate notes.

3 Materials and Methods

3.1 Dataset

JaalTaka [3, 32] (CC BY 4.0) contains 1,390 notes: 802 genuine (398 Tk 500, 404 Tk 1000) from three national banks and 588 counterfeit (320 Tk 500, 268 Tk 1000) supplied by RAB. Each note was photographed six times, showing different segments (Fig. 2), with one of four smartphones (Samsung Galaxy A50 and M31, Redmi Note 8 and Note 9 Pro), giving 8,340 images, which we resized to a longer side of 1,024 pixels. According to the dataset description, 649 original notes (502 genuine, 147 counterfeit) were expanded to 1,390 by augmentation.

3.2 Evaluation protocol

Because the photographs of a note are correlated, we split by note: within each class, notes were shuffled (seed 42) and assigned 70/15/15% to training (562 genuine, 412 counterfeit; 5,844 photographs), validation, and test (120 genuine, 88 counterfeit; 1,248 photographs each). Validation serves model selection and OOD calibration; the test split is used once. Section 4.6 shows that this note-disjoint split is *not* sufficient, because distinct notes can share photographs; we therefore also re-split by duplicate group (Section 4.7). Genuine is the positive class. We report accuracy with a 95% Wilson interval [33], the counterfeit detection rate (specificity), the genuine acceptance rate (recall), and ROC-AUC [34], per photograph (n = 1,248) and per note (n = 208), at threshold 0.5 unless stated otherwise. Because photographs of one note, and notes sharing photographs, are not independent, we also give a group-level bootstrap interval [35]: the 191 duplicate groups (Section 4.6) among the test notes are resampled with replacement 10,000 times, and the 2.5th and 97.5th percentiles of accuracy are reported. Methods are compared on the same test notes with the exact McNemar test [36, 37].

Baselines.

To measure what fine-tuning adds, two baselines use *frozen* ImageNet EfficientNet-B0 embeddings (L2-normalised, same preprocessing as at evaluation). The 1-nearest-neighbour (1-NN) baseline gives each test photograph the label of the training photograph with the highest cosine similarity; the linear probe is a class-balanced logistic regression ($C = 1$) on the training photographs. For both, the note score is the mean of the six photograph scores, as in Eq. (3).

3.3 Classifier and training

EfficientNet-B0 [8], chosen for its accuracy–cost trade-off on CPUs and initialised with ImageNet weights [15], maps a 288×288 input x to a $1280 \times 9 \times 9$ feature map; global average pooling yields the embedding $\mathbf{e} = f(x) \in \mathbb{R}^{1280}$. The head is dropout ($p = 0.3$) and a linear layer:

$$p(x) = \mathrm{softmax}(\mathbf{We} + \mathbf{b})_{\text{genuine}}\,. \qquad (1)$$

Table 1 Scope of prior banknote studies compared with this work (✓: addressed; –: not reported)

Study	Task	Approach	OOD/uncertain	Dupl. audit
[10–12]	Counterfeit	Hand-crafted features	–	–
[13]	Counterfeit (Tk)	Image processing	–	–
[1, 2]	Counterfeit	CNN, smartphone	–	–
[17–19]	Denomination (Tk)	Neural net, CNN	–	–
[3, 25]	Counterfeit (Tk)	CNN, transfer learning	–	–
This work	Counterfeit (Tk)	CNN, multi-view fusion	✓	✓

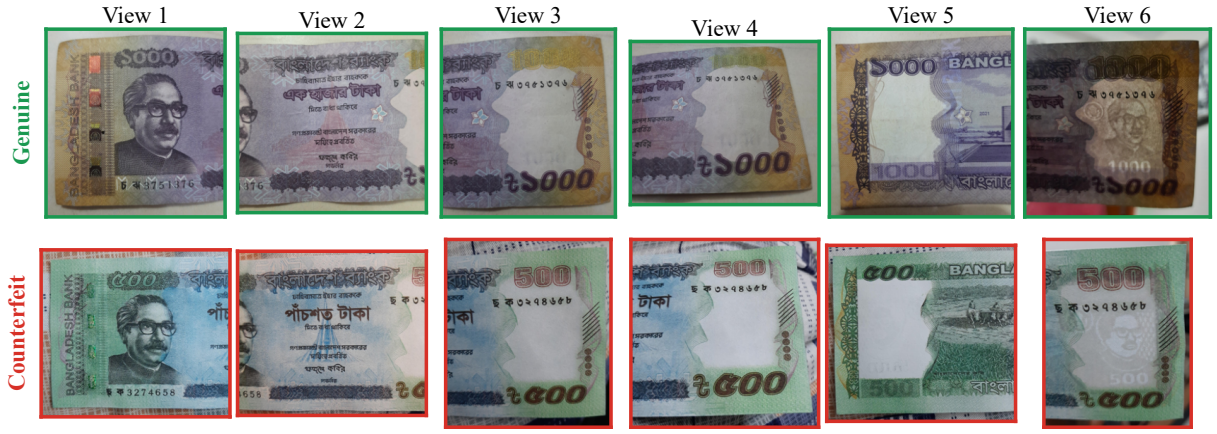


Fig. 2 Six photographs of a genuine Tk 1000 note (top) and a counterfeit Tk 500 note (bottom) from the test split

The model has 4.01 M parameters. Training uses random resized crops (scale 0.35–1.0, aspect 0.75–1.33), rotations up to ±10°, colour jitter (brightness and contrast ±0.4, saturation ±0.3, hue ±0.04), and Gaussian blur ($p = 0.2$). At evaluation, the shorter side is resized to 331 pixels and the central 288 × 288 crop is kept. The stem and first three MBConv stages are frozen with their ImageNet batch-norm statistics; the rest is fine-tuned with the class-weighted, label-smoothed [24] cross-entropy

$$\mathcal{L} = -\sum_{c\in\{0,1\}} w_c\, q_c \log \hat{y}_c,$$
$$q_c = (1-\varepsilon)\,\mathbb{1}[c = y] + \frac{\varepsilon}{2}, \qquad (2)$$

with $\varepsilon = 0.05$ and $w_c = N/(2N_c)$. We use AdamW [38] (weight decay 10^{-4}), a one-cycle schedule [39] (peak 3×10^{-4}, 15% warm-up), batch size 12, and six epochs, keeping the epoch with the highest validation accuracy plus AUC (the sixth: 99.04%, AUC 0.9976). Hyperparameters were fixed beforehand; only the epoch was chosen, on validation data. Training took about 55 min on an Intel Core i5-12400 CPU with PyTorch [40]; all results come from this single run (seed 42), except Section 4.7, where the model is retrained with identical settings.

3.4 Multi-view fusion and three-way decision

Let $\mathcal{V}$ be the photographs of a note that pass the OOD gate. The note-level probability is the mean, a late-fusion form of multi-view pooling [41],

$$\bar{p} = \frac{1}{|\mathcal{V}|}\sum_{i\in\mathcal{V}} p(x_i), \qquad (3)$$

and, for a strictness threshold $\tau \in [0.5, 1)$ (default 0.7), the verdict is

$$\text{verdict} = \begin{cases} \text{not supported} & \text{if } \mathcal{V} = \emptyset, \\ \text{genuine} & \text{if } \bar{p} \geq \tau, \\ \text{counterfeit} & \text{if } \bar{p} < 0.5, \\ \text{uncertain} & \text{otherwise.} \end{cases} \qquad (4)$$

Averaging needs no extra training and accepts any number of views. The asymmetry reflects costs: accepting a counterfeit causes a loss, whereas "uncertain" only asks for a manual check or more photographs.

3.5 Out-of-distribution gate

Following [27], let $\boldsymbol{\mu}$ be the mean training embedding and $\mathbf{U}_k$ its first $k = 64$ principal directions with variances σ_j^2. With $\mathbf{z} = \mathbf{U}_k^\top(\mathbf{e} - \boldsymbol{\mu})$, we compute the in-subspace Mahalanobis distance and the out-of-subspace residual

$$m(\mathbf{e}) = \sum_{j=1}^{k} \frac{z_j^2}{\sigma_j^2}, \qquad \rho(\mathbf{e}) = \|(\mathbf{e} - \boldsymbol{\mu}) - \mathbf{U}_k\mathbf{z}\|_2^2 . \tag{5}$$

Thresholds θ_m and θ_ρ are the 99.5th validation percentiles. A photograph is accepted when $r(\mathbf{e}) = \max\{m/\theta_m,\ \rho/\theta_\rho\} \le 1$ and its Laplacian variance (a texture measure) is at least 30, which rejects blank and blurred images. The residual catches embeddings that leave the banknote subspace, which the in-subspace distance alone cannot detect.

3.6 Explanation and deployment

Grad-CAM [9] is computed on the last feature map $A \in \mathbb{R}^{1280\times 9\times 9}$ for the predicted class c:

$$\alpha_k = \frac{1}{81}\sum_{u,v}\frac{\partial y^c}{\partial A^k_{uv}}, \qquad L^c = \mathrm{ReLU}\Big(\sum_k \alpha_k A^k\Big), \tag{6}$$

where y^c is the logit of class c; the map is normalised, upsampled, and overlaid on the photograph. The model runs in 37 ms per photograph (median of 30 runs, six CPU threads). Reported results use the central crop only.

4 Results

4.1 Main results and baselines

Per note, the system rejects 86 of 88 counterfeits and accepts 118 of 120 genuine notes (Table 2, Fig. 3), and both ROC curves in Fig. 4 rise almost vertically. The per-note confusion matrix equals the per-photograph matrix divided by six, and each view position alone reaches 98.08%: every error comes from all six photographs of one of four notes. Because the errors are clustered, the group-bootstrap interval of the per-photograph accuracy (96.10–99.53%) is about twice as wide as the Wilson interval that treats the 1,248 photographs as independent (97.15–98.70%); the effective sample size is the number of distinct notes, not of photographs.

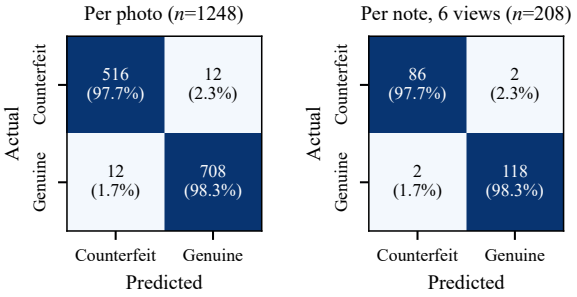


Fig. 3 Test confusion matrices per photograph (left) and per note (right)

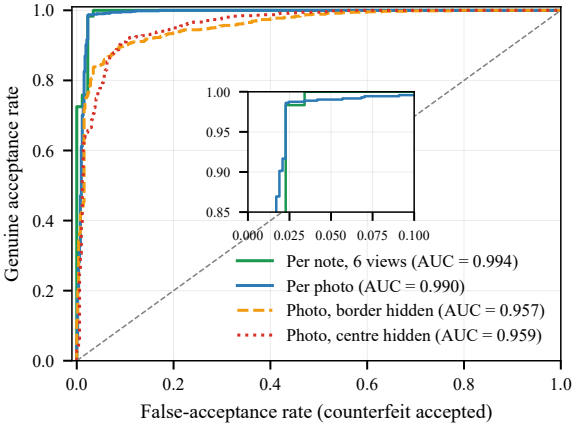


Fig. 4 Test ROC curves for per-note fusion, single photographs, and the occlusion diagnostics of Section 4.5

Against the frozen-feature baselines of Section 3.2 (Table 3), fine-tuning adds 1.4–1.6 points of per-photograph accuracy, but at note level the probe matches it and 1-NN errs on only six notes. Neither note-level difference is significant: the model is right and 1-NN wrong on two notes, never the reverse (exact McNemar $p = 0.50$), and the model and the probe err on the same four notes ($p = 1.0$). The JaalTaka authors' results (97.4–98.9% [3]) used a different split and are not directly comparable; we did not retrain their five CNNs, because Section 4.6

Table 2 Test results of the proposed system (threshold 0.5)

Metric	Per photograph	Per note (6 views)
Samples (genuine / counterfeit)	1,248 (720 / 528)	208 (120 / 88)
Accuracy (%)	98.08	98.08
95% Wilson CI (%)	97.15–98.70	95.16–99.25
95% group-bootstrap CI (%)	96.10–99.53	96.10–99.53
Counterfeit detection (%)	97.73	97.73 (86/88)
Genuine acceptance (%)	98.33	98.33 (118/120)
Balanced accuracy (%)	98.03	98.03
F1, genuine class	0.983	0.983
ROC-AUC	0.990	0.994

shows that the remaining errors are label noise that no architecture can resolve.

4.2 Fusion and the strictness threshold

Over all subsets of k views, accuracy stays at 98.08% while the mean AUC rises monotonically from 0.9900 ($k = 1$) to 0.9939 ($k = 6$): fusion makes the ranking more reliable. The fused scores are strongly bimodal (Fig. 5): correctly classified genuine notes have $\bar{p} > 0.86$ and counterfeits $\bar{p} < 0.10$, so outcomes are unchanged for $\tau \leq 0.85$, and at $\tau = 0.90$ one genuine note (0.8%) becomes "uncertain". The two accepted counterfeits ($\bar{p} \approx 0.95$) would need $\tau > 0.954$, which would send most genuine notes to manual inspection.

4.3 Out-of-distribution gate

The gate rejects 4 of 1,248 test photographs (0.32%) versus 0.72% on validation, with a median test ratio of 0.26 (Fig. 6); no test note was "not supported". In a preliminary check, it rejected 100 of 105 photographs of other denominations (Tk 2–200) and blank, noise, and unrelated images; a systematic OOD benchmark remains future work.

4.4 Visual explanations

In Fig. 7, Grad-CAM concentrates on the portrait and the denomination and ornament panels for correct decisions, and also on the security band for the counterfeit in row b. The wrongly accepted note in row c looks genuine to the model, as expected because an identical copy appears in training labelled genuine (Section 4.6).

4.5 Shortcut diagnostic

As shown in Fig. 8, we blacked out either the centre (keeping a 20% border) or the border of each test input. Hiding the border gives 88.94% accuracy (AUC 0.957). Hiding the centre gives 75.48%, as most genuine photographs are rejected (acceptance 58.47%), yet the AUC remains 0.959 and 98.67% of counterfeits are recognised. The border, which holds the note's margins but often also the table or hand beneath it, thus suffices to rank most photographs. Because genuine and counterfeit notes came from different sources (banks and RAB), capture conditions may contribute to decisions [4].

4.6 Benchmark audit: duplicates and label conflicts

The perfect agreement between a note's photographs, the strong frozen baselines, and the documented augmentation led us to search for duplicates. For every pair of notes, we computed the normalised cross-correlation (NCC) of 48×48 grey-scale thumbnails of corresponding views, averaged over the six views and maximised over flips. The best-match distribution is bimodal (Fig. 9a): for validation and test notes, no match lies between 0.963 and 0.997. We therefore grouped notes with NCC > 0.98 (Table 4). The grouping is stable: any threshold from 0.95 to 0.99 yields 843–846 groups, the same 12 label-conflict groups, and 116–118 test notes with a copy in training or validation. At 0.90 the groups merge to 709, closer to the 649 original notes in the dataset description. Some augmented copies therefore differ more than a near-identical photograph, and our counts are a *lower* bound on duplication.

Table 3 Comparison with frozen-feature baselines on the test split

	Per photograph		Per note	
Method	Acc. (%)	AUC	Acc. (%)	AUC
1-NN, frozen ImageNet features	96.71	0.967	97.12	0.979
Linear probe, frozen features	96.47	0.981	98.08	0.983
Fine-tuned EfficientNet-B0 (ours)	**98.08**	**0.990**	**98.08**	**0.994**

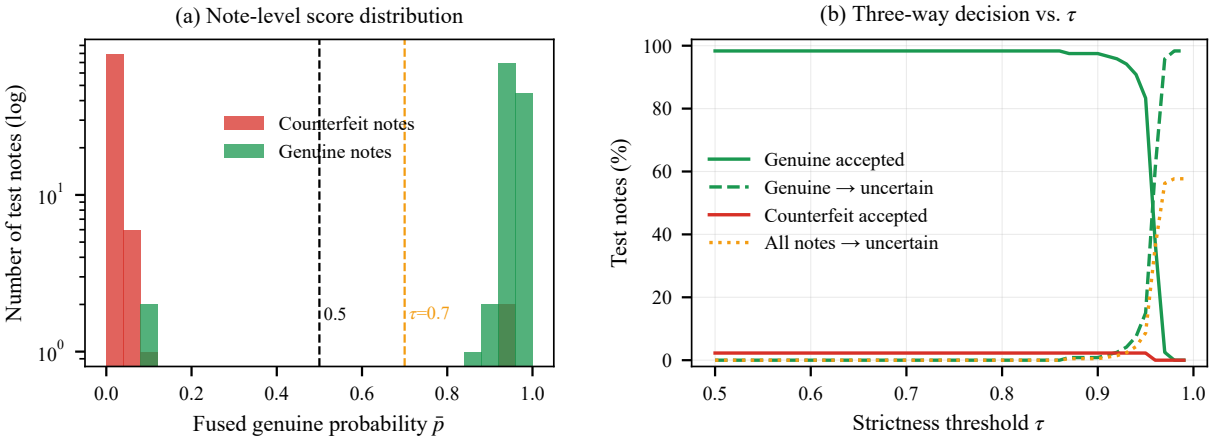


Fig. 5 **a** Fused genuine probability $\bar{p}$ of the 208 test notes. **b** Share of test notes accepted, "uncertain", or wrongly accepted versus the threshold τ

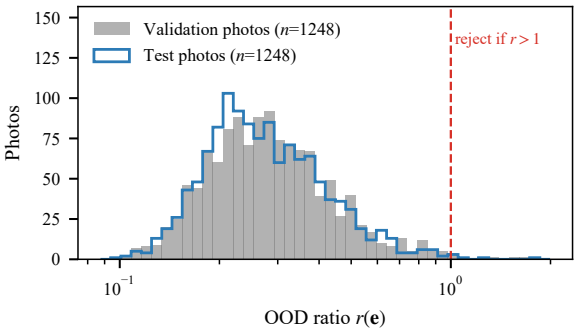


Fig. 6 OOD ratio $r(\mathbf{e})$ of validation and test photographs; the dashed line is the rejection limit

Table 4 Duplicate audit of JaalTaka (pixel NCC > 0.98 over six views)

Quantity	Value
Notes / distinct photograph sets	1,390 / 846
Groups of size 1 / 2 / 3	449 / 250 / 147
Notes sharing photographs with another	941 (67.7%)
Counterfeit-only / genuine-only groups	289 / 545
Groups containing *both* labels	12 (31 notes)
Test notes with a copy in train/val	116/208 (55.8%)
counterfeit / genuine	60/88 / 56/120
Test notes in label-conflict groups	6
Test errors due to label conflicts	4 of 4

The augmented copies were thus stored as independent notes, and a note-level split places copies of 55.8% of test notes in training or validation. Twelve groups contain genuine- and counterfeit-labelled copies of the *same* photographs (e.g., genuine notes 400–406 duplicate counterfeit notes 106–112 and 547–553, and counterfeit notes 400 and 403–406 duplicate genuine notes 802 and 1–4), so at least one label per group is wrong; genuine-labelled notes 402 and 403 carry serial number 1843834, which also appears on counterfeit-labelled notes. *All four* test errors are notes from these groups, each predicted with the label of its training copy. On the 92 *clean* test notes, whose group has no member in training or validation, and on the 202 notes outside label-conflict groups, all three methods are error-free (Table 6). The 98.08% accuracy is thus limited by label noise, not by the model, and the task is saturated even for 1-NN on generic features.

(a) Genuine note, correctly accepted (true: genuine, fused $\bar{p}$=0.972)



View 1 | Grad-CAM, p=0.979 | View 2 | Grad-CAM, p=0.962

(b) Counterfeit note, correctly rejected (true: counterfeit, fused $\bar{p}$=0.007)



View 1 | Grad-CAM, p=0.007 | View 2 | Grad-CAM, p=0.007

(c) Test error: label-conflict note (true: counterfeit, fused $\bar{p}$=0.954)



View 1 | Grad-CAM, p=0.955 | View 2 | Grad-CAM, p=0.951

Fig. 7 Grad-CAM on test notes (views 1 and 2). **a** Genuine, accepted. **b** Counterfeit, rejected. **c** Counterfeit, wrongly accepted; its photographs occur in training under the genuine label

(a) Full input (b) Centre hidden (c) Border hidden



(d) Per-photo test results under occlusion

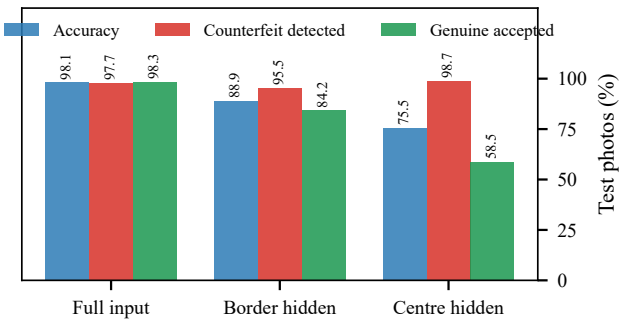


Fig. 8 Occlusion diagnostic. **a**–**c** Inputs with the centre or the 20% border hidden. **d** Per-photograph test results under each condition

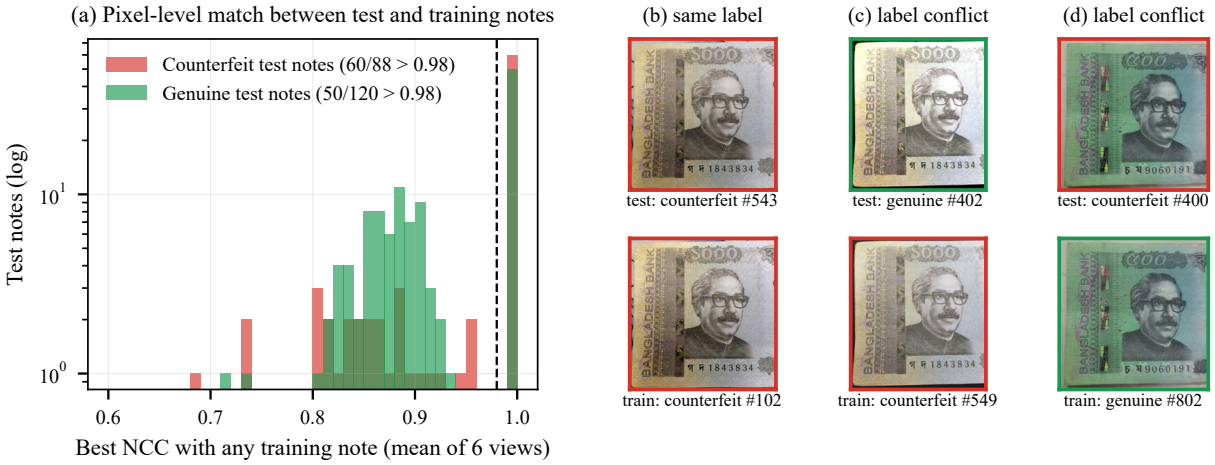


Fig. 9 Duplicate audit. **a** Best NCC match of each test note to any training note; dashed line: threshold 0.98. **b**–**d** A test note (top) and its matching training note (bottom); green frame: genuine, red: counterfeit. **b** Augmented copy, same label. **c**, **d** Identical photographs with opposite labels

4.7 Re-evaluation on a group-aware split

To remove the leakage, we re-split the data by duplicate group: the 31 notes of the 12 label-conflict groups were excluded, and the remaining groups were shuffled within each class (seed 42) and assigned to training (949 notes in 584 groups), validation (205 notes, 124 groups), and test (205 notes, 126 groups: 119 genuine, 86 counterfeit), so that no photograph set occurs in two splits [31]. Table 5 shows that the task does not become harder. 1-NN on frozen ImageNet features classifies all 205 test notes correctly (95% Wilson CI 98.16–100%), although none of them has a copy in training, and the linear probe errs on three notes. Our model, retrained on this split with the settings of Section 3.3 (selected epoch: the third; validation accuracy 99.51%, AUC 0.995), is error-free on all 1,230 test photographs (AUC 1.000); its advantage over the probe is not significant (exact McNemar $p = 0.25$). With the centre of each photograph hidden as in Section 4.5, it still reaches an AUC of 0.999 (accuracy 87.72%), so the border alone almost perfectly separates the classes. Distinct notes of the same class are thus still close to each other in a generic feature space, consistent with shared counterfeit batches and capture sessions rather than with learned security features (Section 4.5). A group-aware split removes copies but not this correlation; only holding out whole counterfeit families and devices can.

5 Discussion

Within JaalTaka, the system is error-free once label conflicts are removed (lower 95% bound 96.0% on the 92 clean notes), its errors are fully explained, and it runs in real time on a CPU. This does not show that it will detect *new* counterfeits: the 588 counterfeits collapse into 289 photograph sets, those we inspected share a few serial numbers, and they were collected separately from the genuine notes. A model, even 1-NN, can therefore succeed by recognising known batches or capture sessions, as the occlusion test suggests. The saturated benchmark also cannot rank methods, since differences between published CNNs are smaller than its label noise.

These findings answer RQ1. The 97.4–98.9% of the original CNNs [3] and our 98.08% all lie within the range that label noise allows, and paired tests find no difference between fine-tuning and generic features. As with near-duplicates in CIFAR [6] and other leakage [7], accuracy partly measures recall of copies; removing them (Section 4.7) does not make the task harder, so the remaining signal separates batches and sessions, not necessarily genuine from forged security features. For RQ2, the system meets (i)–(iii) at 37 ms per photograph on a CPU, and the three-way decision and OOD gate cost no in-distribution accuracy; whether it generalises to new counterfeits cannot be decided with JaalTaka.

Table 5 Test results on the group-aware split (205 notes, no shared photograph sets)

	Per photograph		Per note	
Method	Acc. (%)	AUC	Acc. (%) [95% Wilson CI]	AUC
1-NN, frozen ImageNet features	99.84	0.998	100.0 [98.16–100]	1.000
Linear probe, frozen features	97.15	0.995	98.54 [95.79–99.50]	1.000
Fine-tuned EfficientNet-B0 (ours)	**100.0**	**1.000**	**100.0** [98.16–100]	**1.000**

Table 6 Note-level accuracy (%) on audited test subsets

Test subset	All	Without label conflicts	Clean (no copy)
Notes (genuine / counterfeit)	208 (120/88)	202 (117/85)	92 (64/28)
1-NN, frozen features	97.12	100.0	100.0
Linear probe, frozen	98.08	100.0	100.0
Ours (fine-tuned)	98.08	100.0	100.0
95% Wilson CI	95.16–99.25	98.13–100	95.99–100

We recommend (i) removing or relabelling the 12 conflict groups; (ii) splitting by duplicate group, for which we release our list; (iii) holding out counterfeit families (e.g., by serial number or seizure batch) and capture devices, to measure generalisation to unseen forgeries and phones; and (iv) reporting confidence intervals and frozen-feature baselines, which immediately reveal saturation.

5.1 Limitations

Scope. The system covers only Tk 500 and Tk 1000. Visible-light photographs cannot capture ultraviolet, magnetic, or tactile features, so the output is decision support and suspicious notes should be confirmed at a bank. Scans and distant photographs are mostly rejected, and the data come from four phone models only.

Internal validity. Each model was trained once per split (seed 42); we did not measure variation across seeds. The validation split shares duplicates with training, so the selected epoch and the OOD thresholds were calibrated on partly leaked data. The OOD evaluation is preliminary (105 images, no comparison with other OOD scores), and probabilities are not calibrated on independent data [30]. Grad-CAM maps are shown for selected notes and were not evaluated quantitatively.

Audit validity. Pixel NCC on thumbnails can miss copies changed by strong cropping or geometric augmentation (Section 4.6), so the reported leakage is a lower bound. We cannot tell which label is wrong in the 12 conflict groups without the physical notes, and we compared with the original CNNs only through their published numbers.

6 Conclusion

Our audit shows that JaalTaka, the public benchmark for Bangladeshi counterfeit detection, is saturated: its 1,390 notes form 846 distinct photograph sets, label conflicts explain every test error, and nearest-neighbour classification on generic features is as error-free as our fine-tuned model on duplicate-free test notes and on a group-aware split. Accuracies near 98% therefore cannot rank methods or predict performance on new counterfeits. Our CPU-real-time system with multi-view fusion, a three-way decision, an OOD gate, and Grad-CAM reaches 98.08% note-level accuracy (AUC 0.994). Progress now needs data that separates counterfeit families, sessions, and devices; we release the duplicate groups and splits to support it.

Acknowledgements. The authors thank the Department of Software Engineering, Daffodil International University, for its support and the JaalTaka creators for releasing their data.

Declarations

- **Funding:** This research received no external funding.
- **Competing interests:** The authors declare no competing interests.
- **Ethics approval and consent to participate:** Not applicable. This study involved no human participants, personal data, or animals; it used only banknote photographs from a public CC BY 4.0 dataset [32].
- **Consent for publication:** Not applicable.
- **Data availability:** JaalTaka is available from Mendeley Data [32]. The duplicate groups, the note-level and group-aware split assignments, and the per-note test predictions are provided as Supplementary Information with this article.
- **Materials availability:** Not applicable.
- **Code availability:** The training, evaluation, and audit scripts are provided as Supplementary Information with this article; the trained model weights are available from the corresponding author.
- **Author contributions:** D. Biswas: conceptualisation, software, experiments, analysis, and original draft. I. Mahmud: supervision, methodology review, and manuscript editing.
- **Use of generative AI:** An AI assistant (Claude, Anthropic) was used to help write analysis scripts and to copy-edit the text. The authors designed the study, reviewed and ran all code, verified every result and reference, and take full responsibility for the content.

References

[1] Pham, T.D., Park, C., Nguyen, D.T., Batchuluun, G., Park, K.R.: Deep learning-based fake-banknote detection for the visually impaired people using visible-light images captured by smartphone cameras. IEEE Access **8**, 63144–63161 (2020) https://doi.org/10.1109/ACCESS.2020.2984019

[2] Pham, T.D., Lee, Y.W., Park, C., Park, K.R.: Deep learning-based detection of fake multinational banknotes in a cross-dataset environment utilizing smartphone cameras for assisting visually impaired individuals. Mathematics **10**(9), 1616 (2022) https://doi.org/10.3390/math10091616

[3] Piyas, B.R.C., Tisha, F.J., Rahman, S., Islam, S., Preenon, B.R.C.: JaalTaka: A benchmark dataset for detecting counterfeit Bangladeshi banknotes. Data in Brief **64**, 112377 (2026) https://doi.org/10.1016/j.dib.2025.112377

[4] Geirhos, R., Jacobsen, J.-H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., Wichmann, F.A.: Shortcut learning in deep neural networks. Nature Machine Intelligence **2**(11), 665–673 (2020) https://doi.org/10.1038/s42256-020-00257-z

[5] Kaufman, S., Rosset, S., Perlich, C., Stitelman, O.: Leakage in data mining: Formulation, detection, and avoidance. ACM Transactions on Knowledge Discovery from Data **6**(4), 1–21 (2012) https://doi.org/10.1145/2382577.2382579

[6] Barz, B., Denzler, J.: Do we train on test data? Purging CIFAR of near-duplicates. Journal of Imaging **6**(6), 41 (2020) https://doi.org/10.3390/jimaging6060041

[7] Kapoor, S., Narayanan, A.: Leakage and the reproducibility crisis in machine-learning-based science. Patterns **4**(9), 100804 (2023) https://doi.org/10.1016/j.patter.2023.100804

[8] Tan, M., Le, Q.V.: EfficientNet: Rethinking model scaling for convolutional neural networks. In: Proc. 36th Int. Conf. Machine Learning (ICML). Proceedings of Machine Learning Research, vol. 97, pp. 6105–6114. Long Beach, CA, USA (2019)

[9] Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D.: Grad-CAM: Visual explanations from deep networks via gradient-based localization. In: Proc. IEEE Int. Conf. Computer Vision (ICCV), Venice, Italy, pp. 618–626 (2017). https://doi.org/10.1109/ICCV.2017.74

[10] Yeh, C.-Y., Su, W.-P., Lee, S.-J.: Employing multiple-kernel support vector machines for counterfeit banknote recognition. Applied Soft Computing **11**(1), 1439–1447 (2011) https://doi.org/10.1016/j.asoc.2010.04.015

[11] Berenguel, A., Ramos Terrades, O., Lladós, J., Cañero, C.: Banknote counterfeit detection through background texture printing analysis. In: Proc. 12th IAPR Workshop

on Document Analysis Systems (DAS), Santorini, Greece, pp. 66–71 (2016). https://doi.org/10.1109/DAS.2016.34

[12] Baek, S., Choi, E., Baek, Y., Lee, C.: Detection of counterfeit banknotes using multispectral images. Digital Signal Processing **78**, 294–304 (2018) https://doi.org/10.1016/j.dsp.2018.03.015

[13] Uddin, M.S., Das, P.P., Roney, M.S.A.: Image-based approach for the detection of counterfeit banknotes of Bangladesh. In: Proc. 5th Int. Conf. Informatics, Electronics and Vision (ICIEV), Dhaka, Bangladesh, pp. 1067–1072 (2016). https://doi.org/10.1109/ICIEV.2016.7760162

[14] Dittimi, T.V., Suen, C.Y.: Mobile app for detection of counterfeit banknotes. In: Advances in Artificial Intelligence (Canadian AI 2018). Lecture Notes in Computer Science, vol. 10832, pp. 156–168. Springer, Cham, Switzerland (2018). https://doi.org/10.1007/978-3-319-89656-4_13

[15] Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., Fei-Fei, L.: ImageNet: A large-scale hierarchical image database. In: Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), Miami, FL, USA, pp. 248–255 (2009). https://doi.org/10.1109/CVPR.2009.5206848

[16] Yosinski, J., Clune, J., Bengio, Y., Lipson, H.: How transferable are features in deep neural networks? In: Advances in Neural Information Processing Systems (NeurIPS), vol. 27, pp. 3320–3328 (2014)

[17] Jahangir, N., Chowdhury, A.R.: Bangladeshi banknote recognition by neural network with axis symmetrical masks. In: Proc. 10th Int. Conf. Computer and Information Technology (ICCIT), Dhaka, Bangladesh, pp. 1–5 (2007). https://doi.org/10.1109/ICCITECHN.2007.4579423

[18] Linkon, A.H.M., Labib, M.M., Bappy, F.H., Sarker, S., Jannat, M.-E., Islam, M.S.: Deep learning approach combining lightweight CNN architecture with transfer learning: An automatic approach for the detection and recognition of Bangladeshi banknotes. In: Proc. 11th Int. Conf. Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, pp. 214–217 (2020). https://doi.org/10.1109/ICECE51571.2020.9393113

[19] Nuhash, M.N.I., Akter, S.: BanglaTaka: A dataset for classification of Bangladeshi banknotes. Data in Brief **61**, 111853 (2025) https://doi.org/10.1016/j.dib.2025.111853

[20] Huang, G., Liu, Z., Maaten, L., Weinberger, K.Q.: Densely connected convolutional networks. In: Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA, pp. 4700–4708 (2017). https://doi.org/10.1109/CVPR.2017.243

[21] He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, pp. 770–778 (2016). https://doi.org/10.1109/CVPR.2016.90

[22] Simonyan, K., Zisserman, A.: Very deep convolutional networks for large-scale image recognition. In: Proc. Int. Conf. Learning Representations (ICLR), San Diego, CA, USA (2015)

[23] Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., Chen, L.-C.: MobileNetV2: Inverted residuals and linear bottlenecks. In: Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), Salt Lake City, UT, USA, pp. 4510–4520 (2018). https://doi.org/10.1109/CVPR.2018.00474

[24] Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., Wojna, Z.: Rethinking the Inception architecture for computer vision. In: Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, pp. 2818–2826 (2016). https://doi.org/10.1109/CVPR.2016.308

[25] Tabassum, B., Zubayer, M.H., Masum, A.K.M., Farid, D.M.: Counterfeit Bangladeshi banknote detection using transfer learning. In: Proc. 28th Int. Conf. Computer and Information Technology (ICCIT), pp. 4384–4389 (2025). https://doi.org/10.1109/ICCIT68739.2025.11491481

[26] Hendrycks, D., Gimpel, K.: A baseline for detecting misclassified and out-of-distribution examples in neural networks. In: Proc. Int. Conf. Learning Representations (ICLR), Toulon, France (2017)

[27] Lee, K., Lee, K., Lee, H., Shin, J.: A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In: Advances in Neural Information Processing Systems (NeurIPS), vol. 31, pp. 7167–7177 (2018)

[28] Liu, W., Wang, X., Owens, J., Li, Y.: Energy-based out-of-distribution detection. In: Advances in Neural Information Processing Systems (NeurIPS), vol. 33 (2020)

[29] Sun, Y., Ming, Y., Zhu, X., Li, Y.: Out-of-distribution detection with deep nearest neighbors. In: Proc. 39th Int. Conf. Machine Learning (ICML). Proceedings of Machine Learning Research, vol. 162, pp. 20827–20840 (2022)

[30] Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: Proc. 34th Int. Conf. Machine Learning (ICML). Proceedings of Machine Learning Research, vol. 70, pp. 1321–1330. Sydney, Australia (2017)

[31] Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller, W., *et al.*: Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography **40**(8), 913–929 (2017) https://doi.org/10.1111/ecog.02881

[32] Piyas, B.R.C., et al.: JaalTaka: A Benchmark Dataset for Detecting Counterfeit Bangladeshi Banknotes, Version 2. Mendeley Data. Licensed CC BY 4.0 (2025). https://doi.org/10.17632/2m7wk5cy4c.2

[33] Wilson, E.B.: Probable inference, the law of succession, and statistical inference. Journal of the American Statistical Association **22**(158), 209–212 (1927) https://doi.org/10.1080/01621459.1927.10502953

[34] Fawcett, T.: An introduction to ROC analysis. Pattern Recognition Letters **27**(8), 861–874 (2006) https://doi.org/10.1016/j.patrec.2005.10.010

[35] Efron, B., Tibshirani, R.J.: An Introduction to the Bootstrap. Chapman & Hall, New York, NY, USA (1993)

[36] McNemar, Q.: Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika **12**(2), 153–157 (1947) https://doi.org/10.1007/BF02295996

[37] Dietterich, T.G.: Approximate statistical tests for comparing supervised classification learning algorithms. Neural Computation **10**(7), 1895–1923 (1998) https://doi.org/10.1162/089976698300017197

[38] Loshchilov, I., Hutter, F.: Decoupled weight decay regularization. In: Proc. Int. Conf. Learning Representations (ICLR), New Orleans, LA, USA (2019)

[39] Smith, L.N., Topin, N.: Super-convergence: Very fast training of neural networks using large learning rates. In: Proc. SPIE Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications, vol. 11006, p. 1100612 (2019). https://doi.org/10.1117/12.2520589

[40] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., *et al.*: PyTorch: An imperative style, high-performance deep learning library. In: Advances in Neural Information Processing Systems (NeurIPS), vol. 32, pp. 8024–8035 (2019)

[41] Su, H., Maji, S., Kalogerakis, E., Learned-Miller, E.: Multi-view convolutional neural networks for 3D shape recognition. In: Proc. IEEE Int. Conf. Computer Vision (ICCV), Santiago, Chile, pp. 945–953 (2015). https://doi.org/10.1109/ICCV.2015.114